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Interpretability taxonomy: an approach for artificial intelligence developers technique selection

Carter Cousineau[1*] and Rozita Dara[1]

*Correspondence:
Carter Cousineau
cartercousineau@icloud.com
[1]School of Computer Science, Department and Organization, University of Guelph, 50 Stone Rd, Guelph, ON, Canada

Abstract
As AI systems continue to increase their capabilities of performing human tasks there is a growing need to understand how the AI system determined its decisions. Interpretability is a concept in trustworthy AI research that is focused on understanding of the inner workings and decisions that come from the AI system. Our previous research revealed that AI developers lack consistent approaches or tools for implementing interpretability. There has been substantial theoretical interpretability research, yet the development of practical approaches in the form of tools to assist AI developers on interpretability remains underexplored in the research. This paper develops a taxonomy of AI interpretability techniques based on an analysis of the research literature using the survey of surveys method across 70 papers, examining 30 papers from 2019 to 2023. This paper develops a hierarchical taxonomy at a lower-level of abstraction to present and build upon relevant research literature. This paper also provides an approach as a practical tool in the form of decision trees to help AI developers identify and categorize interpretability techniques based on applicability and characteristics. Unlike existing interpretability taxonomies, this study introduces AI developer-oriented sample decision trees to aid in operationalizing the selection of interpretability techniques. This bridges theoretical research with practical implementation for AI developers. Preliminary testing and validation with AI developers was conducted to ensure applicability of the proposed taxonomy and decision trees. Preliminary validation demonstrated conceptual clarity and practical relevance, forming a foundation for future large-scale evaluation. This research contributes to further bridge the gap between AI systems and ensuring the practical implementation of interpretability.

Keywords Artificial intelligence, Data, Trustworthy artificial intelligence, Interpretability, artificial intelligence developers

1 Introduction

Artificial intelligence (AI) systems are capable of mimicking human intelligence in a wide range of tasks. These AI systems extract information and statistical correlations from large amounts of data to provide valuable insights and support humans

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in decision-making [7, 37]. As these AI systems become more heavily integrated into human lives, it becomes essential to understand how these AI systems arrived at their decisions [12]. This is where interpretability comes in. Interpretability focuses on understanding the reasoning behind an AI system's decisions and outputs [12]. Interpretability is a key concept in the field of trustworthy AI research, offering the ability to understand AI systems to support training data, mitigate risks and gain further insights to AI systems [9, 18, 31, 34]. Interpretability has also been recognized in the research literature in helping to identify poor predictions, aid in fairness detection and obfuscation [48], protect over-fitted AI systems and help to build trust in AI systems [12].

Achieving interpretability in AI systems continues to remain a challenge. AI systems are underpinned by machine learning (ML) algorithms [48]. There are two types of AI systems, black box and white box ML algorithms. White box AI systems allow AI stakeholders to see the internal logic and ML algorithms [48]. Black box AI systems are often AI systems that do not have clear internal logic [48]. The opaqueness of black box AI systems presents a significant challenge [48] due to their lack of transparency, especially when critical decisions depend on the AI system's outputs. Complex ML algorithms that have been found to impact trust and hinder AI end users' adoption of the AI system [48].

The considerations of implementing interpretability occur during the design, development and development phases in the AI system lifecycle. AI developers are at the forefront of designing, developing and deploying AI systems [5, 27]. During these phases in the AI system lifecycle, AI developers are faced with critical decisions to make. Some examples of decisions AI developers make include the selection of data sources and ML algorithms, performance evaluation methods and validate if the AI end users experience is effective. These decisions can impact the trustworthiness and transparency of AI systems [27]. Our prior research found that one of the key decisions AI developers often face is selecting the appropriate interpretability technique(s) to apply to the AI system [11]. AI developers are tasked with ensuring AI systems are interpretable and that the reasoning behind the AI system's outputs can be communicated to AI end users [15]. Interpretability is essential for AI developers to be able to understand the AI system's outcomes so they can provide general explanations to AI end users [10].

Although there is a growing body of theoretical research literature on interpretability, a notable research gap exists between theoretical knowledge and practical approaches or tools for implementation [11, 15, 27, 39, 43, 54]. The current research literature is highly abstract, making it difficult to translate interpretability concepts into practice [47]. Since 2019 the substantial increase in interpretability research publications has resulted in a high volume of interpretability research literature [2, 5, 19, 33, 52, 59]. This high volume of research literature is found to cause greater confusion, lack of coherence and practical guidance for AI developers [52].

To address this gap, our research aims to develop an approach for AI developers to aid their selection of appropriate interpretability technique(s) to apply to their AI system. This proposed approach in the form of a tool leverages insights from the field of interpretable AI research using a survey of surveys meta-analysis. This research has three main research objectives:

- Our first research objective is to gain a greater understanding of interpretability to identify the common concepts and techniques.

- Based on our survey of surveys, our second research objective is to categorize the various interpretability concepts and techniques that are considered to be in scope.
- Our third research objective is to develop an interpretability taxonomy and offer an approach in the form of decision trees for organizations and AI developers to build taxonomies that can guide them in their selection process of interpretability techniques to apply to their AI system.
- Our fourth research objective is to conduct preliminary testing and validation with AI developers to test the applicability of the taxonomy categories and sample decision trees.

Our research study is conducted in two phases. In the first phase, we conducted a broad analysis of recent research literature using a survey of surveys approach to gain a greater understanding of interpretability. We identified the common concepts of interpretability techniques from the meta-analysis. In the second phase of our research, we used these concepts to shape the categories of our taxonomy. We utilized the categories of our taxonomy to construct our proposed tool using a decision trees approach [46]. This proposed tool aims to guide AI developers using a series of questions as decision nodes in our sample decision trees to determine which interpretability technique(s) can be applied to their AI system. As a part of the second phase, we conducted preliminary testing to evaluate the applicability of the interpretability taxonomy and decision trees with AI developers. Applicability in the context of this study refers to whether the interpretability taxonomy categories and decision trees align with AI developers' expectations for assessing interpretability, while identifying any missing aspects that could enhance their AI evaluation processes.

This paper's research contributions include providing a greater understanding on interpretability techniques, develop a hierarchical interpretability taxonomy at a lower-level of abstraction and offer an approach as a process for AI developers to use and modify in guiding their selection of interpretability techniques. We anticipate this approach will be generated and modified in future by AI developers. Our research offers structured guidance for AI developers to support the implementation of interpretability. This study's unique contributions provide: (1) This paper synthesizes interpretability literature using a survey of surveys meta-analysis at a lower level of abstraction; (2) This paper translates the findings into an AI developer-focused taxonomy to support selection of interpretability techniques; and (3) This paper conducts preliminary validation with AI developers to examine practical applicability and usability.

The remaining sections of this paper begin with a review of related studies and background, followed by description of our research design. We then present the key results and findings from this research, provide a discussion, highlights the limitations of this research and a conclusion.

2 Related studies

This section provides a high-level background on the related studies highlighting the research gap between trustworthy AI theory and practical implementation. This section also outlines the growing need in the research for approaches to support and ensure practical implementation of trustworthy AI concepts, with an emphasis on interpretability.

2.1 The gap between interpretability theory and practical implementation

The practice of trustworthy AI research hinges on the ability to comprehend and evaluate the predictions and decisions generated by ML algorithms [16]. There have been recent studies that attempt to capture consistency in interpretability terminologies in the form of theoretical taxonomies. Graziani et al.'s [19] interpretability taxonomy focused on creating a global definition for the term itself because there was no unified definition. The study looked at interpretability in the context of terms such as 'intelligible', 'transparent', 'interpretable' and 'reliable'. This study revealed that these terms can acquire a variety of meanings depending on the context in which they are used [19]. This study led to a variety of interpretations across disciplines [19]. Other studies like Eldrandaly et al. [16] provided a comprehensive taxonomy of explainability (XAI) methods highlighting the importance of securing AI systems, particularly deep learning AI systems, against adversarial attacks. The study conducted a case study with Twitter data sentiment analysis and emphasized the need for AI systems that are both interpretable and secure [16]. The findings highlighted above by Graziani et al. [19] and Eldrandaly et al. [16] grounds the scope of our research highlighting the need for creating a proposed approach in the form of a tool to aid in implementation.

Several recent researchers have explored the theoretical aspects of interpretability. Arrieta et al.'s [5] study provided an in-depth description on the various ML algorithms in AI systems and outlined the details of interpretability techniques. Rai (2020)'s study on the black box and white box concepts provided a detailed overview of the nuances between AI systems and the interpretability techniques [5, 48]. Schwable and Finzel's [52] study conducted a systemic broad literature analysis of XAI research using a survey of surveys approach. This study provided a unified structure to help researchers and practitioners in understanding XAI methods, while also highlighting the importance of context-sensitive application and future research directions in the field [52]. These studies helped inform the structure of our research design and offered clarity on interpretability concepts. By integrating existing taxonomies and theoretical frameworks, our proposed approach can provide a foundational approach to interpretability selection for AI applications.

2.2 The need for practical approaches for implementation

The gap between trustworthy AI theoretical concepts and practical implementation is recognized by various researchers [15, 27, 39, 43, 47, 54, 55]. Morley et al.'s [43] research aimed to better understand if the tools were helping AI developers and stakeholders in implementing trustworthy AI from theory to practice. This study found the current landscape of tools and techniques are either 'too strict' or 'too flexible' for AI developers and stakeholders to use in the form of practical implementation [43]. Eitel-Porter [15] study used a qualitative analysis on the risks and presents a governance framework. Eitel-Porter [15] also outlined the need for mandated governance controls and tools to manage and scale AI systems in organizations. Das and Agarwal's [12] study is a comparative analysis of open-source interpretability tools to gain a greater understanding of ML algorithms behaviours, this study provided value to our research in comprehending the input and output variables and decision-making processes of AI systems. Prem [47] conducted a literature review of 100 ethical AI articles, identified a broad spectrum of tools, approaches and techniques. The tools and techniques ranged from frameworks for

datasets to ML algorithms. This study highlighted the need for further development in areas of labels, best practices and democratic oversight, suggesting that more comprehensive and context-sensitive approaches are necessary for implementing trustworthy AI systems [47].

Our comprehensive meta-analysis outlined in Sect. 3.4 provides a detailed landscape of interpretability and highlighted the key concepts found from our survey of surveys research literature review. The following Sect. 3.4 will describe our research design and methodologies further. The concepts were at a lower-level of abstraction and spanned from dimension metrics that gauge interpretability in an AI system, to the variances of human to computer interaction in AI systems.

2.3 Methods and research design

Our research design used a survey of surveys approach in order to achieve our research objectives, our research design is broken into two phases. As displayed in Fig. 1, Phase 1 is a broad review of interpretability research literature. The survey of surveys is our primary source of paper collection used for the meta-analysis. We analyzed the survey of surveys to gather the common concepts. The concepts from this analysis defined and categorized the layers of our hierarchical taxonomy. Phase 2 of our research developed an approach by using the root nodes and decisions nodes from the interpretability categories outlined in the survey of surveys. We used the root nodes and decisions nodes to shape the taxonomy layers in the form of decision trees and fine-tuned the proposed interpretability taxonomy. Afterwards, we took our interpretability taxonomy and sample decision trees to conduct a preliminary test with the AI developers.

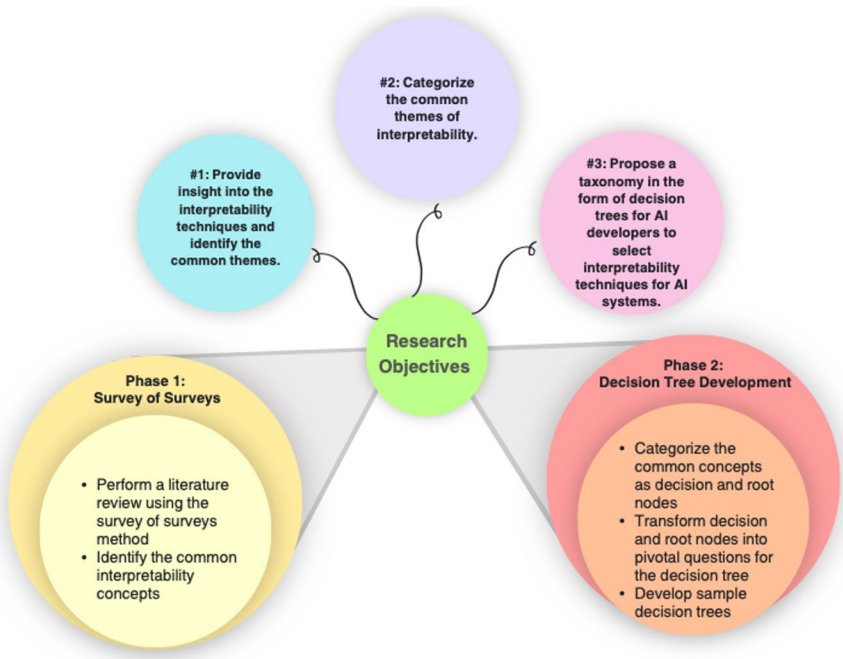

Fig. 1 High-level research design

Table 1 Search phrases used for defining literature review scope

Search phrases
Interpretable AI Taxonomy
Interpretable AI Taxonomy Proposed Tool
Interpretable AI Taxonomy Tool for AI Developers to Select Interpretability Techniques

Table 2 Inclusion and exclusion criteria

Criteria	Inclusion	Exclusion
Year of Publication	2019 to 2023	2018 and under
Title of paper	A minimum of 2 of the following key terms: 'interpretable', 'taxonomy', 'proposed', 'tool', 'guide', 'AI developers', 'developers', 'AI', 'techniques'	Does not have 2 of the inclusion key terms
Abstracts	The abstract displays relevancy to our research objectives.	The abstract did not display relevancy to our research objectives.
Most frequently cited	More than 50 citations	Less than 49 citations

2.3.1 Phase 1: survey of surveys literature review

Our research centers on interpretable taxonomies, with a focus on developing a tool as an approach for selecting interpretability techniques to apply to AI systems. Inspired by Schwable and Finzel's [52] our research design used similar methodology and instead uses a lower-level of abstraction on the concepts from the survey of surveys to identify and determine the categories for our interpretability taxonomy. A lower level of abstraction is valuable because it enables a deeper understanding of foundational concepts and methodologies, allowing for more precise analysis. Our survey of surveys included research literature published from 2019 to 2023. The target meta-surveys included interpretability taxonomies and theoretical frameworks to determine the meta-characteristics for our taxonomy [46, 56].

2.3.2 Search, inclusion and exclusion criteria

We first analyzed the existing research taxonomies and theoretical frameworks in the field of interpretability. Our literature review search was broken into two sub-phases. Sub-Phase 1.1 of the search focused on interpretability taxonomies (as outlined in Table 1) and Sub-Phase 1.2 of our search used additional inclusion and exclusion criteria (as outlined in Table 2) to narrow the scope of research literature in the survey of surveys.

2.3.3 Search phase 1.1: interpretability taxonomies

Prior to conducting our search of relevant research literature, we identified common terms with stemming to frame the basis of our search:

- AI: artificial intelligence, machine learning (ML) and algorithms.
- Interpretability: interpretable and interpret.
- Taxonomy: taxonomy, taxonomies, guide, tool, toolbox, toolkit.

We then applied the search phrases in Table 2 into Google Scholar and counted the number of research publications that were in the search category. As noted in Table 2, this process allowed us to narrow our scope to only include the relevant research publications.

2.3.4 Search phase 1.2: inclusion criteria

We developed additional inclusion and exclusion criteria as outlined in Table 3 to narrow our scope of literature papers used in the survey of surveys. We filtered our search to the publication years ranging from 2019 to 2023. We prioritized the articles that were under the 'reviewed' category in Google Scholar and met the inclusion and exclusion criteria in Table 3. We then did a scan of the titles of the papers and abstracts to ensure relevancy. It is through the combination of Table 3's inclusion and exclusion criteria that allowed us to have a defined scope of research papers. We also included relevant surveys that were pointed out by the reviewers, totalling 30 surveys examined.

The survey of surveys approach draws from relevant prior studies. In order to enhance inclusion, multiple iterative rounds of literature review and reviewer input were conducted to mitigate the risk of exclusion. To reduce potential selection bias, 2 reviewers independently screened papers across multiple publishers and verified inclusion decisions to ensure coverage breadth and relevance. As a part of this process, certain AI systems categories such as ML algorithm types, such as bayesian, GAN, reinforcement learning, hybrid, and large language model (LLM) systems were excluded due to their limited and inconsistent representation in meta-survey literature between 2019 and 2023, which precluded systematic comparison. This scope ensures methodological consistency, it limits generalizability to emerging AI systems. Future work should extend the taxonomy to these AI systems.

The following Sect. 3.5 will go over our findings. The surveyed papers and supplementary materials used in this study were sourced from the following publishers: Elsevier Science Direct, Wiley, IEE Xplore, SpringerLink, MDPI, nature machine learning, ACM Digital Library, arXiv Cornell University, emerald insights, The Royal Society Publishing, and the bmj. An important distinction is that preprints were used solely as supplementary materials to complement peer-reviewed literature, with all sources critically evaluated and peer-reviewed works prioritized to ensure academic rigor. Table 3 provides a summary of the papers in our survey of surveys, identifying the year of publication, number of papers in that year and the range of volume of citations.

2.4 Phase 2

2.4.1 Developing a hierarchical taxonomy

With our survey of surveys papers reviewed, we identified the common concepts as the categories for our hierarchical taxonomy. Hierarchical taxonomy is an iterative taxonomy method that begins with determining the meta-characteristic and progresses through cycles of empirical to conceptual and conceptual to empirical iterations to refine the taxonomy [46, 56]. We used this methodology to systemically develop a hierarchical taxonomy and ensure each level is validated and refined through various iterations [46, 56]. In order to develop a hierarchical taxonomy that had practical use, we

Table 3 Summary of the survey of survey papers

Year of publication	# of Papers	Range of # of citations
2023	4	57 to 116
2022	5	105 to 485
2021	8	61 to 1352
2020	7	177 to 7033
2019	6	367 to 3248

used decision trees to shape the structure of our approach. Decision trees are often used in sequential models to classify and combine logically sequences of tests [28]. We used decision trees to support our structured approach to categorizing the key concepts using an empirical to conceptual and conceptual to empirical method [46, 56]. In the context of our research, decision trees are used to derive the interpretability categories of our taxonomy in the form of root and decisions nodes. Our decision trees are shaped using the root nodes, decision nodes and leaf nodes. Figures 12, 13 and 14 are our sample decision trees and outline the root nodes, decision nodes and leaf nodes. Root nodes build the decision tree [28] and define the set path that the tree goes through (outlined in Fig 6. labelled as A). Decision nodes are the points at which the path reaches a decision in the form of a question to be answered (outlined in Figs. 12, 13 and 14 labelled as B). Decision nodes continue down the path to the next decision node associated to the node [28]. Finally, leaf nodes are the final determinant down the path (outlined in Figs. 4, 13 and 14 labelled as C). We had 3 AI experts review the sample approaches and provide annotation that has been incorporated into this paper.

With an understanding of our methodologies used, the following Sect. 3.5 details the results from our survey of surveys. As the foundation of the results stem from the survey of surveys, Sect. 3.5, specifically 3.5.1 to 3.5.14 outlines the design and development of our proposed interpretability taxonomy providing two sample decision trees.

Once the interpretability taxonomy and sample decision trees were developed in Phase 2, we conducted preliminary testing and validation of the taxonomy and sample decision trees with AI developers. The methodologies used for the preliminary testing and validation are outlined further in Sect. 3.4.6. Section 3.5 identifying our interpretability taxonomy, sample decision trees and preliminary testing.

2.4.2 Preliminary testing and validation of taxonomy and sample decision tree

The purpose of the preliminary testing and validation is to determine if the interpretability taxonomy and sample decision trees can act as a starting point in providing a greater understanding of interpretability and offer support to AI developers. Our preliminary testing focused on:

1. Assess and test the applicability of the interpretability taxonomy categories and sample decision trees (Figs. 4 and 13) to determine whether they align with AI developers' expectations for evaluating interpretability.
2. Identify missing aspects or details that could enhance applicability for AI developers.

2.4.3 Participants selection and sampling procedure

AI developers in the context of this research are defined as individuals who are responsible for designing, developing and/or deploying AI systems [11, 15]. For the purpose of this study, AI developers needed to possess some familiarity with interpretability and also have some experience designing, developing and/or deploying AI systems. As shown in Table 7, we recruited industry practitioners in AI, data science, and information technology as our participants using a random sampling approach. In order to ensure we conducted random sampling; we reached out to a variety of AI developers from our prior study. We also used Amazon Mechanical Turk's audience to collect feedback from a larger group of AI experts and increase our target sample size (12 participants). Of the 12 participants in this study: 3 participants were AI developers from a prior study, 3

participants were recruited through professional network and 6 participants were Amazon Mechanical Turk's audience. Table 7 is a breakdown of the participant demographics that varied in age and gender. Our study had a total of 12 participants (outlined in Appendix C Table 7), each participant reviewed and provided informed consent. A purposive sample of 12 AI developers were selected to obtain qualitative insights. The findings demonstrate preliminary relevance, justifying value in future empirical validation studies.

We conducted a mixed-methods research approach with both closed and open-ended questions to capture participants quantitative metrics and qualitative insights. Each participants' data points were converted to csv format for data analysis. In order to validate the aspects of interpretability, we provided each participant in the study with the following: (A) The details of the interpretability taxonomy outlining the 8 categories, followed by (B) A set of semi-structured questions for participants to answer. The validation was intentionally limited to a small group of AI practitioners to prioritize qualitative findings. This design supports conceptual refinement and informs future large-scale validation.

2.4.4 Analytics approach

Once the data was collected; we cleansed and combined the data into a single file for analysis. We used the grounded theory method for thematic analysis from the data collected. Grounded theory method takes qualitative feedback to draw thematic analysis from the data [4, 22, 32, 58]. The coding categories were extracted through an iterative process where the themes emerged from the data based on recurring concepts and relationships (as displayed in Table 4). Table 4 displays a short summary of the coding categories alongside a short description that we used to derive our results. The following Sect. 3.5 will provide details of our findings.

3 Results

Before we delve into the results section, it is important to share the operational definitions that we use throughout this Chapter. We define AI systems as a broad concept encompassing various approaches, including ML, expert systems and natural language processing (NLP). Our research prioritizes ML and NLP, as they present greater interpretability challenges, whereas expert systems rely on predefined rules that are generally more transparent.

3.1 Interpretability survey of surveys

We reviewed over 30 surveys, the survey of surveys method was employed to gather a sub-selection of data using the most relevant surveys [52]. These 30 papers were chosen for analysis. We used the common concepts found across these surveys as our method to categorize interpretability concepts, create the hierarchical taxonomy and offer

Table 4 Coding categories

Coding category	Description
Systematic approach and clarity *(results in Sect. 3.5.15.1)*	Ensuring interpretability and comprehensive coverage of concepts.
Some examples of practical application *(results in Sect. 3.5.15.2)*	Providing practical examples and use cases to better understand the possible usability by AI developers.
Structure and coverage *(results in Sect. 3.5.15.3)*	Structuring information effectively for the taxonomy categories that are useful and support AI developers.

approaches in the form of decision trees. Where additional context was required, we expanded our review of the research literature to incorporate other relevant papers and supplementary material published between 2006 and 2024. Figure 2 is comprised of the interpretability concepts from our survey of surveys results outlined in Sub-Sect. 3.5.2 to 3.5.14.

3.1.1 The concept of interpretability

As Fan et al. [17] and Graziani et al. [19] pointed out there is a lack of consensus in the research literature on the definition of interpretability, leading to the diversity in interpretability. Interpretability can be categorized in 3 different levels: simulatability, decomposability and algorithmic transparency [5, 17]. Simulatability involves understanding the entire AI system (outlined in Sect. 3.5.4) [5, 17]. Decomposability involves analyzing the components within AI systems, this can include the data types, number of features and layers (outlined in Sect. 3.5.5) [5, 17]. Finally, algorithmic transparency involves understanding the AI system training details (outlined in Sect. 3.5.6) [5, 17]. With the meaning of interpretability in mind we analyzed the surveys to determine the common categories.

3.1.2 The relationship between interpretability and XAI

Interpretability and XAI are trustworthy AI concepts that are often used interchangeably in research [1, 19] and used interchangeably in practice by AI developers [11]. It is worth highlighting the important distinction between these two concepts from our meta-analysis and draw the parallel on how these concepts are critically important in conjunction. Interpretability refers to the inherent transparency of a model (further outlined in Table 7) and the specific techniques that allow us to understand how the AI system's inputs influence its outputs [12]. For example, various interpretability techniques are Layer-wise Relevance Propagation (LRP), Shapley Additive Explanations (SHAP) or Local Interpretable Model-Agnostic Explanations (LIME) (further outlined in Sect.

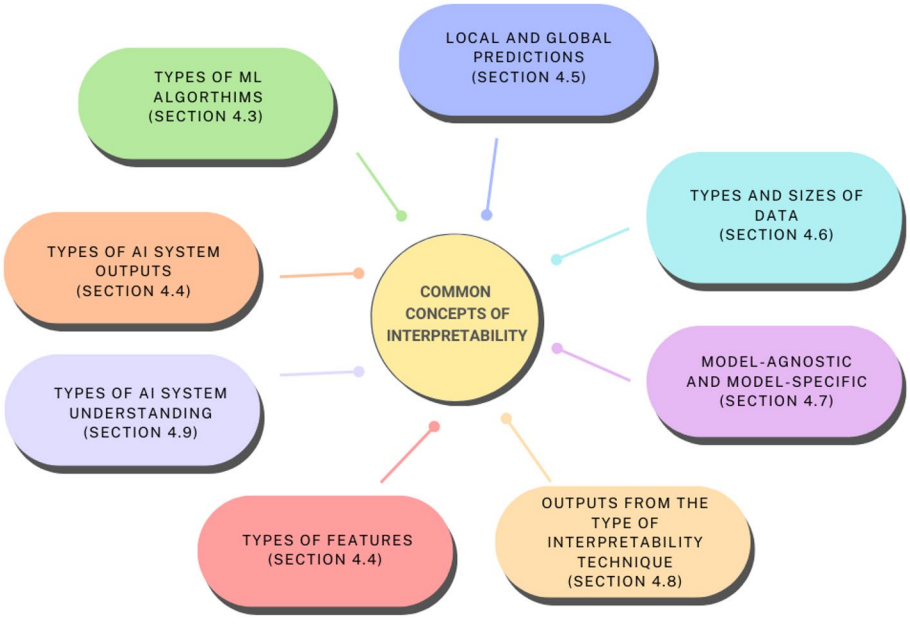

Fig. 2 Common Interpretability Concepts Across the Survey of Surveys

3.5.9). Each interpretability technique aims to provide detailed insights into the decision-making of AI systems, making the AI system transparent and easier to understand [12]. Whereas XAI is a broader framework that encompasses interpretability techniques as one of its core components [37]. In essence, interpretability techniques are specific tools within the broader XAI framework and seeks to make AI systems as a whole more understandable and trustworthy [52, 56].

3.2 Taxonomy category 1: simulatability: types of ML algorithms

AI systems are comprised of various types of ML algorithms. ML algorithms are the backbone in an AI system and vary in complexity [40, 50]. The first relevant taxonomy category of interpretability that we found in the surveys was the importance of understanding the various types of ML algorithms. A visual illustration of some of the various types of ML algorithms that can make up an AI system is included in Appendix A. Our illustration highlights the differences in each type of ML algorithm to provide an understanding of how the AI system derives outputs and decisions. As seen in Appendix A, Fig. 6 helps to visually display how AI system's decisions and outputs are made. Appendix B summarizes descriptions of the various types of ML algorithms across literature.

As defined in Sect. 3.5.2, there is an important distinction to acknowledge from Arrieta et al.'s [5] study. Some of the ML algorithms that are outlined further in Sect. 3.5.3 are inherently interpretable and transparent by nature. Table 5 details the ML algorithm types that fall into the inherently interpretable with simulatability, decomposability and algorithmic transparency. There is a trade-off between the ML algorithms performance and its ability to be interpreted [38]. The disparities among ML algorithm types emphasized the necessity for a distinct taxonomy category as a decision node in our proposed decision trees.

The findings underscore a persistent tension between interpretability and AI system performance. Transparent AI systems often enhance trust but may reduce predictive accuracy, whereas complex AI systems can obscure reasoning despite higher performance. These trade-offs highlight the ethical dimensions of interpretability, particularly regarding bias detection, fairness, and accountability. Embedding interpretability early in AI system design can help balance transparency with technical performance.

Table 5 A summary of arrieta et al.'s [5] table of XAI of ML algorithms

ML algorithm type	Inherent simulatability, decomposability, algorithmic transparency
Rule based	√
Decision trees	√
Ensembles	
Linear and logistic regression	√
K-nearest (KNN)	√
Support vector machines (SVM)	
Bayesian	√
Neural network (includes RNNs and CNNs)	
Reinforcement learning	
Natural language processing (NLP)	
Generative adversarial networks (GAN)	√

3.3 Taxonomy category 2 and 3: decomposition: types of outputs and features

There was not only a variation of ML algorithms that can be used in AI systems, but AI systems can also provide a variety of outputs [51]. Features are individual data characteristics in an AI system [60]. For instance, a feature can be how often a term is used throughout a piece of text or a feature can also be a user's engagement with a product. These features are critical to help the AI system determine patterns [47]. Interpretable features are described as understandable to humans or generated by humans [60]. The second relevant taxonomy category of interpretability found in the survey of surveys was the variation of AI system outputs and features. Zytek et al. [60] identified the most useful features in an AI system have meaning to AI end users. For instance, by understanding and summarizing the features that are relevant to the AI end user's work [60]. The larger the number of features or predictions, the greater the complexity of interpreting the AI system [60]. When the AI system is going to be used by AI end users, the importance of interpretable features was found essential to aid in decision making [60].

3.4 Taxonomy category 4: algorithmic transparency: local or global predictions

The fourth relevant taxonomy category in interpretability is the type of predictions. Our survey of surveys revealed that offering a general understanding of the AI system enhances AI end users' trust [50]. Local predictions attempt to make complex AI systems interpretable through a single prediction, using a subset of the data [13, 14, 53]. Local predictions are single predictions in the AI system [6]. Whereas global predictions are made from the larger data source(s) or usage pattern(s) in the AI system.

Global predictions derive a high-level understanding of the AI system [6]. Arya et al.'s [6] research highlighted AI end users in their study sought to have a static general understanding of the AI system. Interpretability is intended to provide alignment for the AI end user's mental framing of how the AI system operates [24]. Thus, our approach in the form of decision trees will use interpretability with global predictions techniques to provide AI end users a general understanding of the AI system.

3.5 Taxonomy category 5: types and sizes of data

The survey of surveys found that the scale of data source(s) can impact the choice of interpretability techniques [14]. The fifth key category in the interpretability taxonomy emphasizes understanding the specific data sources used in the AI system and how the ML algorithms utilize this data as input to generate predictions or decisions. This category highlights the need to interpret not only the data itself but also the ways in which algorithms process and leverage it within the system [16]. Aldoseri et al. [3] provided a detailed overview examining data source(s) in the AI system, highlighting there are challenges in data quality, volume, size and types. Data can be in various forms, either structured or unstructured. Structured data typically refers to organized and labeled data, commonly found in numerical or categorical data [51]. Unstructured data lacks clear organization or labeling, often in the form of text, audio, sequences, images or video files [51]. Understanding the data is a key requirement not only for selecting interpretability techniques but also for AI end users to be able to consider using the AI system [7]. If an AI system used small data source(s) the AI developer may select local interpretable model-agnostic (LIME) or shapley additive propagation (SHAP) due to their ability to provide insights at instance levels [36, 50]. The size and complexity of the data sources

within an AI system necessitate careful consideration of interpretability requirements [20].

3.6 Taxonomy category 6: model-agnostic or model-specific

The sixth relevant taxonomy category of interpretability is the distinction between model-agnostic and model-specific techniques. Model-agnostic techniques are designed to be applied to the AI system regardless of the AI system's internal workings and the techniques can be used on a variety of ML algorithms [5, 52]. While model-specific techniques are tailored to explain the specific type of ML algorithms and require access to the inner workings of the AI system to generate explanations that are specifically for the ML algorithm [5]. This is an important distinction in our interpretability meta-analysis as Arrieta et al.'s [5] taxonomy illustrated the high-level relationship between post-hoc explainability, specifically model-agnostic and model-specific techniques to the types of ML algorithms. Post-hoc explainability are techniques that attempt to describe the AI system's behaviour from the ML algorithm with a single prediction [16, 23]. There is a large body of research literature on post-hoc methods [5, 35]. Arrieta et al.'s [5] overview found various types of ML algorithms can be applied to various interpretability techniques, emphasizing the variety of interpretability techniques that AI developers are faced with selecting.

3.7 Taxonomy category 7: outputs based on the type of interpretability technique

As highlighted in Sect. 3.5.1 to 3.5.8, the research literature captured an array of categories surrounding interpretability techniques adding further variation to an AI developer's decision making for what interpretability technique to apply. For the purposes of our research, we list the more common interpretability techniques found from our research analysis, noting out review was not all encompassing we prioritized the most commonly referenced interpretability techniques for this paper. The following is a summary of some of the techniques for AI interpretability we reviewed:

- *Decisions or Rule Extractions* These techniques extract a set of rules or decisions derived from the features to determine the outputs of the AI system. It clarifies how specific features contribute to the decision-making process [8].
- *Layer-wise Propagation (LRP)* This technique calculates relevance scores at each layer of the AI system. These scores propagate through the network, helping to identify how each feature contributes to the final output [41].
- *Sensitivity Analysis* This technique assesses how changes in specific features affect the outputs of the AI system. By analyzing these changes, it determines the sensitivity and importance of each feature to the system's performance [21].
- *Image-based Explanations* These techniques generate heat maps and provide spatial visualizations highlighting the regions of an image that.
- The AI system focuses on during classification [57]. The heat maps help to identify the image elements most influential in determining the system's outputs [57].
- *Permutation Feature Importance* This technique that evaluates the importance of each feature by shuffling the feature values and observing the effect on the AI system's performance. A decrease in performance indicates the significance of the

shuffled feature, providing insights into how the model learns patterns for its outputs [25].

- *Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME)* These techniques are commonly used for model-agnostic interpretability. SHAP uses a game theory approach to determine the contribution of each feature to the output, while LIME approximates the model locally to explain predictions [49].
- *Partial Dependence Plots (PDP)* This technique visualizes the effect of specific features on the AI system's output by averaging the predictions across a range of feature values. This technique helps in understanding how changes in a feature influence the overall model prediction [42].
- *Surrogate Models* This technique employs an interpretable ML algorithm to mimic the behaviour of a more complex AI system, making it easier to understand [26]. Surrogate model is a technique that may be used to explain black-box complex AI systems [30].

3.8 Taxonomy category 8: types of AI system understanding

In turn, an important delineation was the variety in the types of understanding that interpretability techniques can provide about the AI system. Understanding the various types of interpretability required by different AI stakeholders is crucial for the successful adoption and acceptance of AI systems. Different AI stakeholders, ranging from domain experts, regulatory bodies to AI end-users, have diverse needs when it comes to understanding AI decisions [5]. For example, global predictions offer a high-level understanding of the AI system that can be in the form of spatial visualizations, non-linear or categorical relationships, changes in feature values to impact predictions or features and predicted outcomes [5]. These explanations may be better suited for AI developers instead of AI end users. Thus, it is essential to tailor explanations to meet the varying needs of the AI stakeholders to build trust, ensure transparency and facilitate the responsible use of AI across different contexts [16]. For example, logistic regression AI systems may be easily understood and thus can get organizational buy-in because the AI system is somewhat trustworthy [44]. Whereas an AI system with more complex ML algorithms require alternative interpretability techniques to be able to explain and interpret the AI system's decisions [44]. As Hong et al.'s [24] participants found the explanations of an AI system to be more frequently requested when the AI system is interfacing with AI end users. The interpretability of an AI system was found to provide decision making for AI end users interfacing with the AI system. Hong et al.'s [24] also found that providing interpretability was an important sign of 'due diligence' as means to further trust and comfort. As Ribeiro et al. [50] pointed out there is a need to provide AI end users an understanding of the AI system's behaviours and a rationale for why the AI system's outputs are made. Ribeiro et al. [50] argues model-agnostic interpretability techniques are generic to allow for flexibility in the choice of explanation from the AI system. With these parameters in mind, we limited our scope to the interpretability techniques that offer global predictions as the starting point.

3.9 Common interpretability concepts from the survey of surveys

Interpretability techniques can be used on an array of AI systems in summary here are the key concepts found from our survey of surveys:

- *Taxonomy Category 1* Types of ML Algorithms (Sect. 3.5.4).
- *Taxonomy Category 2* Types of AI System Outputs (Sect. 3.5.5).
- *Taxonomy Category 3* Types of Features (Sect. 3.5.5).
- *Taxonomy Category 4* Global or Local Predictions (Sect. 3.5.6).
- *Taxonomy Category 5* Types and Sizes of Data (Sect. 3.5.7).
- *Taxonomy Category 6* Model-specific or Model-agnostic (Sect. 3.5.8).
- *Taxonomy Category 7* Outputs Based on the Type of Interpretability Technique (Sect. 3.5.9).
- *Taxonomy Category 8* Types of AI System Understanding (Sect. 3.5.10).

These categories must be considered in combination for AI developers to choose appropriately.

Narayanan and Shmatikov's [45] work found the complexity of the AI system impact the efficiency and AI end user satisfaction. The complexity of the explanation was defined by its size, number of repeated terms and cognitive chunks [44]. It is important to also consider these categories in parallel because interpretability decisions often involve multiple overlapping factors, such as the complexity of an AI system and the time it takes to execute an interpretability technique (as highlighted in Sect. 3.5.15.3). Balancing these trade-offs ensures that AI systems achieve the right level of interpretability without significantly compromising performance or practicality.

3.10 Inclusion and exclusion criteria for developing our decision trees

Before we defined the root and decision nodes for our approach, we developed a set inclusion and exclusion criteria, in Table 6. The exclusions were selected based on the evidence in the survey of surveys to provide sufficient clarity on the interpretability techniques and help narrow down which interpretability techniques can be applied. To narrow the scope of our approach, our initial scope only focused on model-agnostic, global interpretability techniques as the starting point.

Table 6 Inclusion and exclusion criteria

Criteria	Inclusion	Exclusion
ML algorithms	Rule-Based, Decisions Trees, Reinforcement Learning, Ensembles (Random Forests), Linear or Logistic Regressions, KNN, SVMs, Neural Networks (RNNs, CNNs, Multi-Neural Networks), Rule-Based, NLP	Bayesian, GANs, Hybrid Systems
Audience for Using the proposed tool	AI developers	All other AI stakeholders
Audience for the outputs of the proposed Tool	AI end users of the AI system	
Model-agnostic and model-specific	Model-agnostic	Model-specific
Predictions	Global predictions	Local predictions

3.11 Defining the root and decision nodes

We classified the interpretability techniques based on the results from Sect. 3.51–3.5.10, following the categories outlined in Sect. 3.5.11. These categories were selected to capture the key interpretability concepts, drawing from our survey of surveys results. Our taxonomy considers the lower-level abstractions of: the ML algorithm, prediction type (local or global), feature complexity, data size, data type, model approach (agnostic or specific), and desired explanation outcome. The following are the categories of our interpretability taxonomy, provide a high-level definition of the taxonomy category and share the affiliated question-based nodes for each taxonomy category:

1. *Types of ML Algorithms* This taxonomy category is defined as each type of ML algorithm (as outlined in Sect. 3.5.4).

- *The types of sample questions for this category may include* Does the AI system only use one type of ML algorithms in the AI system architecture? Which one of the following ML algorithms were used? Is the AI system using a single decision tree or more than 1?

2. *Global or Local Predictions* This taxonomy category is defined as local and global predictions (as outlined in Sect. 3.5.6).

- *The types of sample questions for this category may include* Is the interpretability technique, global and/or local predictions? Is the goal to have local or global interpretability? Are you looking for a single prediction?

3. *Types of Features* This taxonomy category is defined as the number of features (as outlined in Sect. 3.5.5).

- *The types of sample questions for this category may include* Are you interested in understanding a certain feature? Are you primarily concerned with understanding the changes in input features? Do you need to measure the sensitivity of the AI system's features values? Are you looking for feature level explanations across the AI system?

4. *Model-specific or Model-agnostic* This taxonomy category is defined as identifying whether the explanation is looking for a model-specific of model-agnostic approach (as outlined in Sect. 3.5.8).

- *The types of sample questions for this category may include* Are there specific features that are most influential compared to other features? Does each layer of the neural network contribute to the final outputs of the AI system? Do you require the average affect across features?

5. *Types and Sizes of Data* This taxonomy category is defined as the size of the data source(s) (as outlined in Sect. 3.5.7).

- *The types of sample questions for this category may include* What is the total size of the data used in the AI system? What types of data were used in the AI system? Is the data in the AI system using text or sequences only? Do you want to understand the spatial visualizations?

6. *Types of Interpretability Techniques* This taxonomy category uses the various interpretability techniques in research (as outlined in Sect. 3.5.9).

- *The types of sample questions for this category may include* Which model techniques apply to your AI system? Are you interested in understanding the feature and predicted outcome?

7. *Types of AI System Outputs* This taxonomy category is defined as the various types of outputs the AI system can create (as outlined in 3.5.5).

- *The types of sample questions for this category may include* Are you looking for instance-level explanations? Does the AI system make predictions or decisions as outputs? How complex is the AI system? Can the AI system have misinterpreted explanations?

8. *Types of AI System Understanding* This taxonomy category is defined as the high-level understanding of the interested explanation and outcomes (as outlined in Sect. 3.5.10).

- *The types of sample questions for this category may include* What are type of explanations will aid in understanding the AI system? Are you interested in understanding changes in feature values to impact AI predictions? Are you interested in understanding the specific inputs? Are you looking to understand the effect a feature has across the AI system? Would detailed instance-level explanations be preferred?

By structuring the taxonomy as decision trees with question-based nodes, we aim to provide an approach as a tool with ease of use for selecting interpretability techniques. As demonstrated in Sect. 3.5.11 and 3.5.12, the taxonomy can be applied to various AI systems to generate interpretability. We provide two sample outcomes of our proposed taxonomy in Sect. 3.5.14 and 3.5.15. These outcomes were constructed using Lucidchart software [29].

3.12 Approaches using our taxonomy categories in the form of decision trees

Our proposed approach navigates the complex landscape of interpretability. We developed a set of criteria in the form of root and decision nodes as a means to narrow down the options for AI developers to select interpretability techniques. We proposed leveraging decision tree structure to enable the key categories of interpretability as an implementation tool in AI systems. Inspired by Markus et al.'s [38] proposed framework for determining XAI methods based on 3 steps: (1) understanding the importance of performance; (2) the complexity of the model; and (3) the importance of either fidelity or interpretability. We use a similar decision-making process in Fig. 3. The utilization of our proposed approach should not be used as a final product, instead act as a tool to aid in the selection of interpretability techniques that would best apply to the AI system. The scoping of root and decision nodes is to help further narrow the options of interpretability techniques that are applicable to the type of AI systems.

For demonstration purposes, we provide two samples. The first sample used ensembles/random forests ML algorithms (in Fig. 4). The second sample used recurrent neural networks (RNNs) ML algorithms (in Fig. 5). We selected ensembles/random forests ML algorithms as our first sample, as it is common and more transparent type of ML

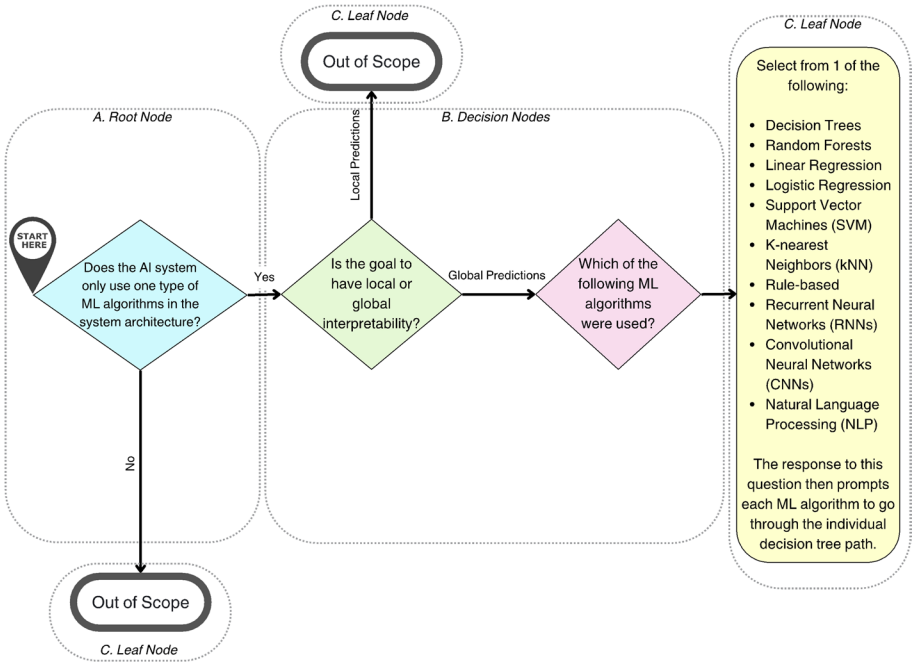

Fig. 3 Scoping and selection criteria

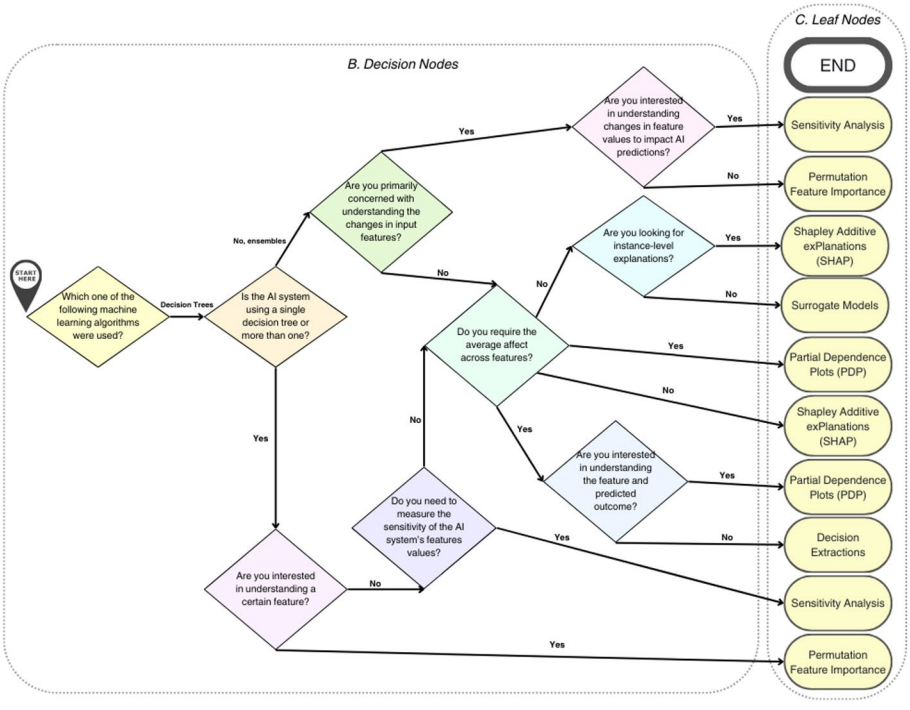

Fig. 4 Sample 1: using the interpretability taxonomy for ensembles algorithms

algorithms in the AI system [5]. As the complexity increases in decision trees the AI system then becomes an ensemble or random forest, we felt showing both in our first sample would demonstrate the selection process for less or more complex ML algorithms. We selected RNN ML algorithms as our second sample as it is a more complex

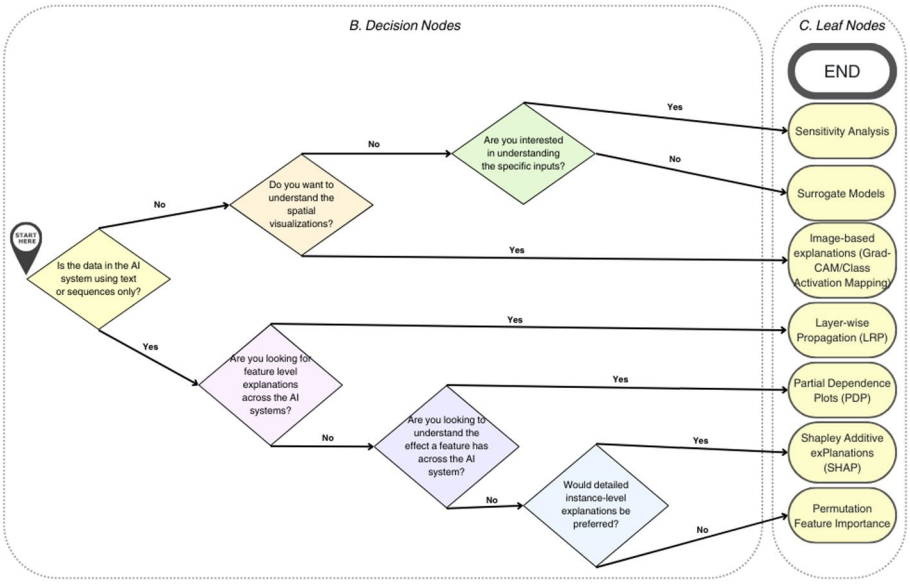

Fig. 5 Sample 2: using the interpretability taxonomy for recurrent neural network (RNN) algorithms

neural network AI system with a variety of possible interpretability techniques that may be applied.

Figure 3 displays our proposed starting scope and selection process. Once the primary source of ML algorithms is selected, the same approach then moves into the distinct decision tree taxonomy path (in the form of questions) based on the type of ML algorithms selected. The tool starts at the 'start here' on Fig. 4, then goes through a series of decision nodes (set of questions). Based on the leaf node from Fig. 3, the path of decision nodes outlined in Figs. 5 and 14. The AI developers are expected to answer these questions and continue through the questions (decision nodes) until they reach the leaf node. The leaf node distinguishes between the various interpretability techniques to apply to the AI system, this includes sensitivity analysis, permutation feature importance, SHAP, surrogate models, PDP and decision extractions.

Unlike the approach in Fig. 4, interpreting RNNs requires nuanced decision nodes (set of questions). Figure 5 illustrates an approach for a decision tree path for RNN ML algorithms. Following a similar approach to ensembles/random forests ML algorithms Fig. 5 starts with the entry point labelled as 'start here' on the left side of Fig. 5 and the AI developers goes through a set of questions to determine the leaf node. AI developers navigate through the root and decision nodes for RNN environments until they reach the recommended interpretability technique. For the purpose of this sample the possible recommended global interpretability techniques for RNN ML algorithms include: sensitivity analysis, surrogate models, grad-CAM and class activation mapping, LRP, PDP, SHAP and permutation feature importance.

Figures 4, 13 and 14 takes the core categories of our taxonomy and displays two sample approaches in the form of decisions trees to apply to various AI systems. The two samples act as a proposed process for AI developers to use when considering why types of interpretability techniques can best apply to their AI system. We note these proposed samples identify a process for how to take the taxonomy into practical implementation for further testing and use by AI developers. This proposed tool aims to support AI developers to narrow down their selection of interpretability techniques that the AI

developer would then still have to apply to the AI system. The sample decision trees were designed as a starting point to use in our preliminary validation study. The sample decision trees are designed to help translate conceptual taxonomy layers into a usable, AI developer-oriented tool.

3.13 Preliminary testing and validation results

3.13.1 Interpretability taxonomy and sample decision tree: systematic approach

The survey of surveys examination highlighted the fact that there is a vast array of interpretability techniques in research literature. We asked participants "How challenging do you find it to select the right interpretability technique for your AI system?". Participants in this study identified the many interpretability techniques make it is difficult for AI developers to determine which interpretability technique to select:

> *Q9: "It's very challenging because new approaches keep coming up and it is hard to know which one applies to my use case."*
> *Q10: "I think this is one of the main sort of core challenges. I would say it's context dependent or sometimes problem specific, but I would say it leaning more towards challenging to like a like an extreme challenge at times only because I feel like it from personal experience I face."*

This correlates to a prior study we conducted with AI developers, wherein we found AI developers identified implementation of interpretability challenging in ensuring appropriate selection and deployment [11].

There were several interesting findings from the response of participants. All participants (100%) in this study had highlighted the importance and value of our structured taxonomy to help them better navigate the complex landscape of interpretability. Participants expressed that our taxonomy categories provided a useful systematic approach to help them know about interpretability methods and select them:

> *Q2: "This taxonomy provides a strong foundation for a structured approach to interpretability. It ensures that all critical aspects are considered, helping me avoid ad hoc decisions and focus on techniques that align with my project's goals."*
> *Q10: "It would definitely prove efficiency. Yes, this taxonomy and decision trees are definitely good." "…having [this interpretability taxonomy] to sort of help narrow down the scope is extremely valuable."*
> *Q3: "The 8 categories and the example provide a solid foundation for a structured approach to achieving interpretability in AI systems. The categories cover a broad range of key factors, such as Types of ML Algorithms, Global vs. Local Predictions, and Types of Features, which are all critical for understanding the context of the model and determining the most appropriate interpretability techniques. This structured breakdown ensures that developers can take a systematic approach, considering various aspects of the model and data before deciding on the best methods for explanation."*
> *Q1: "…The breakdown of categories allows for thoughtful and methodical decision-making, helping ensure consistency and transparency in how interpretability is approached across different AI projects."*

The interpretability categories were found to help avoid inconsistent or unmethodical selection. The participants found our interpretability taxonomy and sample decision trees efficient in providing a solid foundation of what critical aspects should be considered in selecting the interpretability technique for their AI system.

As the number of interpretability techniques can be overwhelming for AI developers, the participants further suggested how the taxonomy and decision trees would simplify the selection process to help narrow down to relevant interpretability techniques:

> *Q7: "It does help in guiding you through the platitude of different techniques that are out there. I think it's a really good way to narrow down your search into what to use for your specific use case… The decision tree is really clear and simple and really helpful."*
> *Q9: "I think because the space of approaches is very large, I need categories that helps me narrow down the area of interest and from what I have seen this would allow me to relatively easily split the possibilities into an area that limits my choice and makes it easy to focus on what I truly care about."*
> *Q3: "The decision tree also adds value by guiding the developer through a series of logical steps to match interpretability techniques to specific needs, such as whether you're explaining global behavior or focusing on instance-level explanations. This guidance ensures a more consistent approach, making it easier to choose techniques that align with project goals and stakeholder requirements."*

This structured approach, facilitated by the taxonomy, was found to help AI developers identify the most suitable interpretability methods for their needs. The taxonomy helped by enabling AI developers to filter out irrelevant techniques, allowing them to focus on those that are most applicable to their AI system and overall application. The participants also emphasized that our approach is appropriate and offers categorization that helps reduce ad hoc decision making and guide AI developers toward the most relevant methods based on the type of AI system, data scale and specific needs.

Participants found the categories to help further distinguish between the various aspects of interpretability, such as global and local interpretability (as described in Sect. 3.5.6), model-specific and model-agnostic techniques (as described in Sect. 3.5.8), as well as various data and feature types:

> *Q8: "The taxonomy effectively covers key elements of interpretability, such as distinguishing between global and local interpretability, model-specific versus model-agnostic techniques, and various types of features, outputs, and data. These categories offer a well-rounded approach for AI developers to consider different interpretability needs based on the model type, data scale, and feature complexity. The inclusion of interpretability techniques and their applicability to different AI systems ensures practical relevance."*
> *Q10: "I think it does help and at least there is a benchmark that speaks to which type of interpretability should be preferred for a specific topic or group type."*

This structured approach was found to help AI developers consider interpretability aspects before making the decision of which technique to apply, leading to more robust and informed choices.

3.13.2 Some examples of practical application

We asked participants if there are examples of how they see themselves using or not using this interpretability taxonomy and sample decision trees. Participants did not have any examples where they felt this could not be used in practice. Instead, participants provided some examples of where they felt this could be used in practice:

> *Q2: "I would use it as a checklist to ensure a thorough and consistent process for achieving interpretability in all my AI projects."*
> *Q3: "For example, knowing whether you're dealing with a model-specific or model-agnostic technique helps guide the decision on which interpretability method to apply."*
> *Q5: "While a sample decision tree would visually represent a simple classification task, clearly demonstrating how features are used to make predictions at each node, allowing for easy interpretation of the decision-making process."*
> *Q10: "In this scenario, it was very difficult to communicate to the C-Suite the difference between why one model would not be a best fit for all their customer segments. We would need multiple models and if there was, for example, if there was [an interpretability process] like this in which we were to communicate."*
> *Q11: "…in a project using neural networks for entity resolution, the taxonomy would guide the selection of model-agnostic techniques like SHAP to highlight feature importance, ensuring both global and local interpretability to meet various user requirements. This structured approach ensures that technical decisions are in line with project objectives and stakeholder expectations."*

This finding identifies that AI developers see value in using the taxonomy to guide their interpretability decisions and structuring their approach. The examples of practical application suggest that the interpretability taxonomy and sample decision trees help validate this systematic approach and can help navigate the complexities of interpretability techniques for AI developers.

3.13.3 Interpretability taxonomy and sample decision trees: structure and coverage

The participants highlighted the structured process as a useful means to help them in their interpretability decision making processes. Some of the AI developers offered a few additional areas of consideration for researchers, such as understanding the relationship between the AI system's complexity, execution time, as well as the trade-off between interpretability and overall performance of the AI system:

> *Q9: "Is there anything else? I mean I don't know if some interpretability techniques take a very long time to execute and our very complex."*
> *Q8: "…the trade-off between interpretability and performance, and ethical considerations surrounding interpretability (e.g., bias detection and fairness explanations)."*
> *Q2: "I think there could be a bit more emphasis on situational flexibility in the guidance. While the categories cover broad cases well, there are sometimes situations where multiple categories might overlap, and selecting interpretability techniques can be more complex. In such cases, having some guidance on how to prioritize different categories or combine methods might enhance the overall structure and offer a more nuanced approach."*

Based on this finding, we refined the overall description of our taxonomy categories (in Sect. 3.5.11) to provide additional context on how our taxonomy categories should be used interchangeably when selecting an interpretability technique for the AI system. Some participants expressed the need for further guidance on navigating trade-offs and prioritizing categories.

Participants also revealed that interpretability decisions are rarely made in isolation; rather, the AI developers' decision involves balancing competing priorities between the AI system's performance, computational complexity, and ethical considerations such as bias and fairness. Several AI developers described how increasing transparency of the AI system can reduce predictive accuracy or slow execution time, prompting difficult trade-offs in production settings. Participants also emphasized that interpretability choices are context-dependent and require situational flexibility, especially when multiple taxonomy categories may overlap. These insights underscore that interpretability is not purely technical challenge, instead it is an ongoing negotiation between the AI system's performance, combination of ethical concepts and practical feasibility.

As interpretability is a rapidly evolving research field, AI developers highlighted the importance of ensuring the taxonomy is consistently updated with emerging techniques, interdisciplinary applications and domain-specific use cases:

> *Q6: "It's essential that the taxonomy evolves with emerging trends in AI, addresses interdisciplinary applications, and provides enough flexibility to apply it across different domains. Decision trees offer an inherently interpretable model, they can become less interpretable as they grow in complexity. Techniques like pruning rule extraction or using decision trees are surrogate models for more complex systems can help retain their value in interpretability. For larger or ensemble models, supplementary techniques like feature importance and global visualization tools can be used to extract useful, understandable insights. Combining the right taxonomy of interpretability techniques with simple models like decision trees or more complex ones ensures that AI systems are both powerful and transparent, helping stakeholders make more informed, responsible decisions."*
> *Q11: "To make the framework more robust, could consider adding … feedback loops and iterative improvement: Integrating processes for ongoing refinement."*
> *Q9: "Especially if this keeps being updated as new techniques comes out, I think this is very helpful."*

AI developers emphasize that the interpretability taxonomy and sample decision trees should be flexible enough to adapt across different industries such as healthcare and finance. This finding aligns to Morley et al.'s [43] trustworthy AI, citing the importance to provide tools and approaches that are flexible.

4 Discussion

Our proposed approach serves as a structured tool for selecting interpretability techniques in AI systems, aiming to enhance interpretability. This discussion highlights key findings and their broader implications.

The survey of surveys analysis identified some strengths and limitations of various interpretability techniques. It also revealed important trade-offs; as interpretability techniques differ in terms of accuracy, complexity and applicability. Understanding these

trade-offs is essential for optimizing interpretability while balancing computational efficiency and applicability.

While prior taxonomies such as Graziani et al. [19], Arrieta et al. [5], Schwable and Finzel [52] provide primarily classification of interpretability concepts. This study extends these efforts by operationalizing taxonomy categories through decision tree logic, providing a practical framework for AI developers. Our hierarchical taxonomy classifies interpretability techniques into root and decision nodes, using lower-level abstractions to provide AI developers with a structured framework for navigating the diverse field of interpretability. Ideally, offering systematic categorization enables AI developers the opportunity to make informed choices when selecting techniques for specific AI systems, reducing ad hoc decision-making.

Additionally, the taxonomy helps to clarify interpretability concepts by distinguishing between model-specific and model-agnostic methods, separating the diverse nature of these techniques. Our preliminary testing found AI developers recognize the value of structured taxonomies in guiding interpretability decisions. By categorizing techniques in a clear and systematic way, the taxonomy improves transparency and reduces uncertainty in selecting interpretability methods. The participants emphasize the taxonomy must remain flexible to incorporate emerging interpretability techniques and variation of AI systems. These interpretability techniques continue to evolve, and AI developers will need to iteratively generate and refine approaches to ensure adaptability in the future.

As AI systems become increasingly integrated into society the importance of ensuring their transparency and reliability is critical. By structuring interpretability techniques through taxonomy and decision trees, our approach facilitates informed decision-making, helping AI developers choose appropriate methods that enhance the transparency, trustworthiness, and usability of AI systems.

5 Limitations in research scope

As the research literature on interpretability is heavily populated, we had to narrow and limit our research scope. The narrow research scope has led to further limitations in the research itself. This study does not include all variations of ML algorithms, Bayesian, GANs, Reinforcement Learning, Hybrid or Large Language Model AI systems are excluded from this study. Our research should only act as a starting point and will require further testing and iterations on general usability across various applications. As well, at present there is limited knowledge on what if any are the benefits from the outputs of using this proposed tool for the AI end users. Thus, these limitations provide additional areas for further research work. It was unclear in the research literature whether certain interpretability technique may work in certain circumstances or applications, a possible area for future research work. The taxonomy currently excludes bayesian, GAN, reinforcement learning, and large language model systems, which may limit applicability to emerging architectures. Subsequent iterations should incorporate these AI paradigms and evaluate ethical considerations such as fairness, explainability, and human-AI accountability to strengthen alignment with trustworthy AI principles.

The interpretability taxonomy in the form of decision trees can be used by researchers for future work and extend into proposed use by organizations. For future work by researchers, it provides a structured approach to design experiments, conduct

comparative studies and advancing interpretability research. It underscores the importance of interdisciplinary research when considering interpretable AI. For organizations, our interpretability taxonomy in the form of decision trees offered guidance to AI developers and other stakeholders in selecting interpretability techniques for AI systems. It emphasized the need for interpretability to be integrated into the design and development process of AI systems to help ensure trust in the AI systems.

The preliminary testing and validation include some insights from AI developers' perspectives. Further research studies can conduct an in depth evaluation of how effectively AI developers can integrate the taxonomy into their decision making processes and examine how it impacts the interpretability outcomes in practice. This research used a small-scale validation with 12 AI developers as an initial step to assess initial conceptual validation and applicability. While the findings provide early evidence of practical relevance, the findings do not represent the full diversity of AI practitioners. Future work should expand validation through larger, domain-specific studies, longitudinal assessments, and comparative experiments to evaluate scalability and impact.

This research revealed gaps in the existing research literature, such as the limited exploration of interpretability in reinforcement learning and the need for future research work in interpretability on complex AI systems, such as hybrid AI systems. By bridging theoretical research and practical implementation, this study contributes a foundational yet adaptable framework that should evolve with new interpretability techniques and guide future development of trustworthy AI systems.

6 Conclusion

The construction of an interpretability taxonomy in the form of decision trees for the selection of interpretability techniques acts as practical tool that contributes to the advancement of AI research and practical implementation. A classification for interpretability techniques added harmonization for AI developers. Our decision tree approach can possibly aid researchers, AI developers, policymakers and other stakeholders in fostering trustworthy AI. This research is a starting point for practical tooling that aims to help ensure AI systems are developed and deployed in a manner that is interpretable.

Appendix A

See Fig. 6.

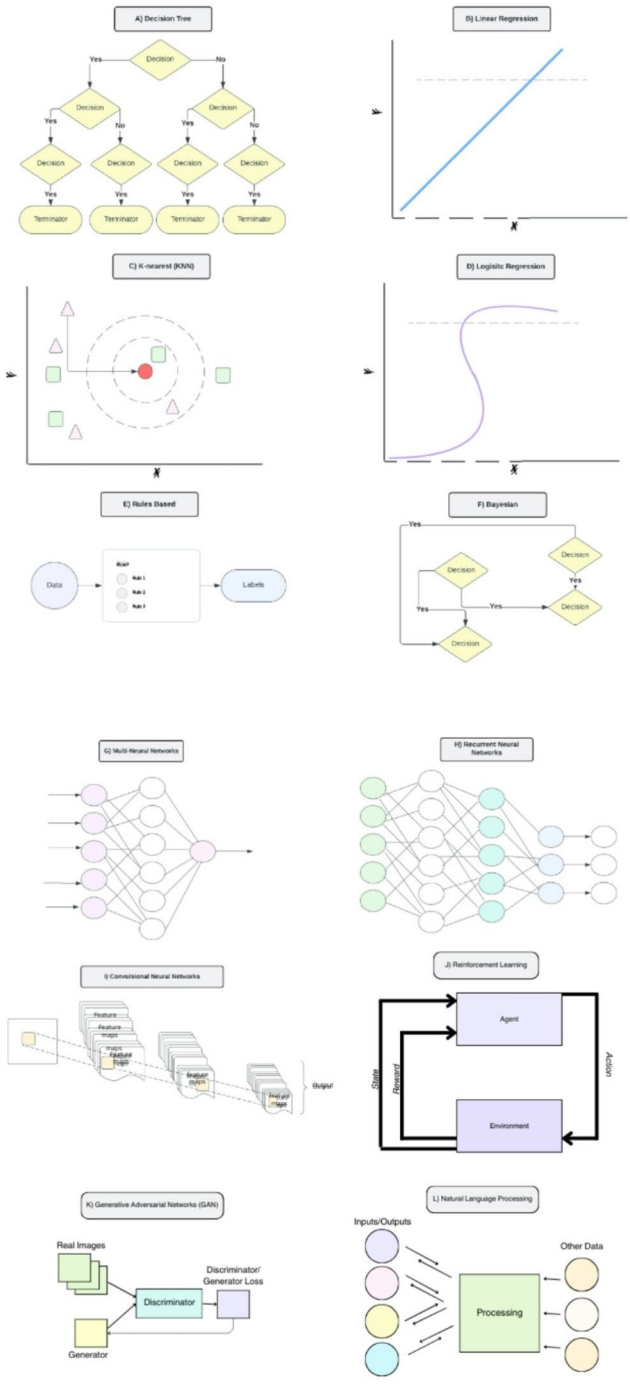

Fig. 6 Visual Illustration of ML Algorithms

Appendix B

Arrieta et al.'s [5] and Fan et al.'s [17] studies described in great length the various types of ML algorithms, as summarized below:

- Rule Based (Fig. 6E) These types of ML algorithms use a defined set of rules in the decision-making and outputs of the AI system [20].

- Decisions Trees Classifiers (Fig. 6A) These ML algorithms function through a series of logical decisions, where nodes represent decisions, branches connect these nodes, and leaves denote outcomes, allowing for retracing the steps to understand the reasoning behind the AI system's outputs [5, 51].
- Ensembles (Fig. 6A) These ML algorithms, commonly known in the research as Random Forests or Gradient Boosting, combine multiple decision trees to achieve higher accuracy in AI system outputs, though they require more monitoring of decision-making features [5].
- Linear or Logistic Regressions (Fig. 6B and D) These types of ML algorithms assign features to each of the data inputs and outputs. Linear Regression utilizes the straight-line pattern, whereas logistic regression utilizes the S-shaped pattern [5].
- K-nearest (KNN) (Fig. 6C) These types of ML algorithms classify in a simple way, wherein it predicts based on the class nearest to its neighbors [5].
- Support Vector Machines (SVMs) These types of ML algorithms have greater complexity than random forests, they construct hyper plane(s) to influence dimensional space [5].
- Bayesian (Fig. 6F) These types of ML algorithms use a probabilistic approach on a set of variables. Bayesian algorithms can convey the clear representation of the relationship of features if the variables are not overly complex [5].
- Neural Networks (Fig. 6G) These ML algorithms can be in the form of Multi-Neural Networks, Recurrent Neural Networks (RNNs) (Fig. 6H) or Convolutional Neural Networks (CNNs) (Fig. 6I). Neural Networks have many layers of inputs throughout the processing of the AI system adding greater complexity in re-tracing the decisions from its outputs [5].
- Reinforcement Learning (Fig. 6J) These ML algorithms learn through their interactions with the environment to enable machines to evaluate the optimal behaviours [51].
- Natural Language Processing (NLP) (Fig. 6)): These ML algorithms are used for an AI system to process and understand natural written or spoken language [51].
- Generative Adversarial Networks (GAN) (Fig. 6K) These ML algorithms are used as a form of deep learning to generate characteristics that are similar to the input data [51].
- An important note is this study does not include all ML algorithms, we excluded further examination on Bayesian, GANs, reinforcement learning, hybrid and large language model AI systems as the research literature surveys was limited. This is an area of opportunity for future research.

Appendix C

See Table 7.

Table 7 Participants demographics

Type	Age	Gender	Participant number
Industry practitioner in AI and data science	25–45	Female	Participants 8, 9
		Male	Participants 7, 10, 11, 12
Industry practitioner in information technology		Female	Participants 3, 4
		Male	Participant 1, 2, 5, 6

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Author contributions
C.C. wrote the main manuscript text, prepared figures and tables. All authors (C.C. and R.D.) reviewed the manuscript.

Data availability
The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

Code availability
Not applicable.

Declarations

Ethics approval and consent to participate
The research in this paper were approved by the University of Guelph's Review Ethics Board (REB), and all procedures followed the ethical guidelines set by the university and relevant regulatory bodies.

Consent for publication
Not applicable.

Informed consent
Consent obtained by each participant and provided informed consent.

Competing interests
The authors declare no competing interests.

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